WHAT LIES BEYOND THE CUT

How much turbulence can a coarse grid know about?

Approaching the optimal closure: equivariance, inductive bias, and Reynolds-number generalization in data-driven LES

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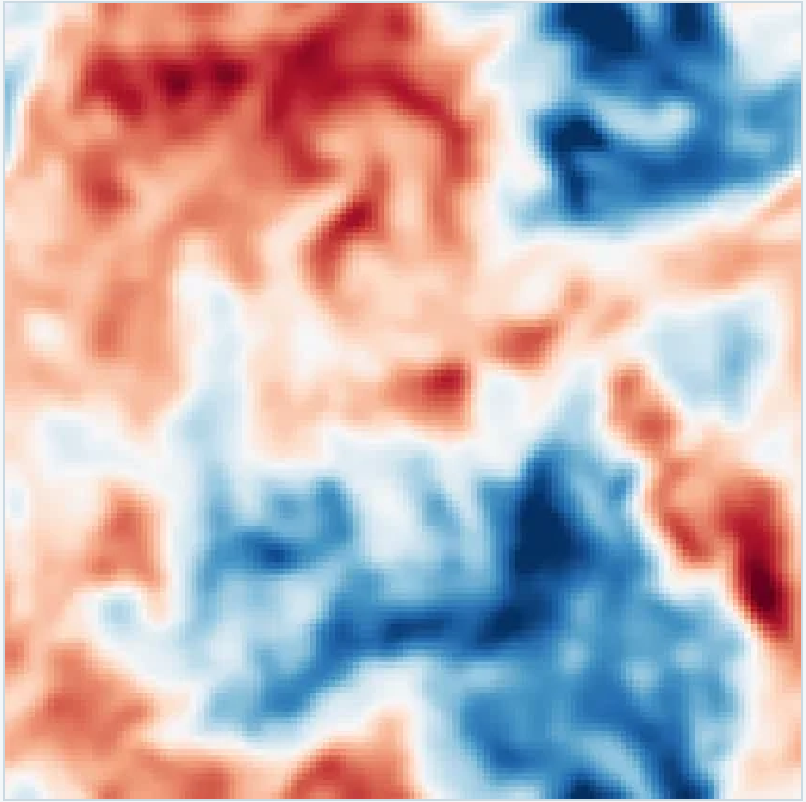
MS184A · Theory-guided Design of Deep Learning-based Surrogates

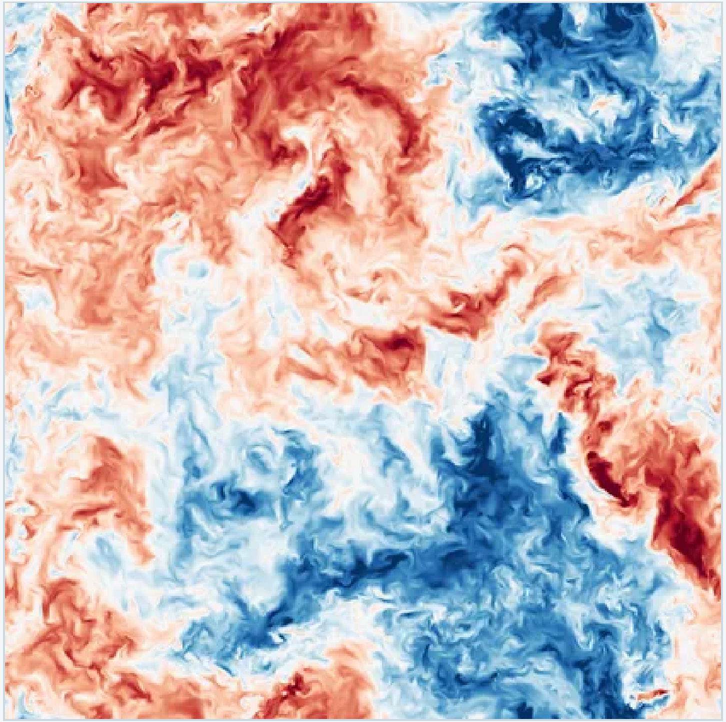
Funded by NWO Talent Programme Vidi (VI.Vidi.193.105)

compute on SURF / Snellius (EINF-15798)



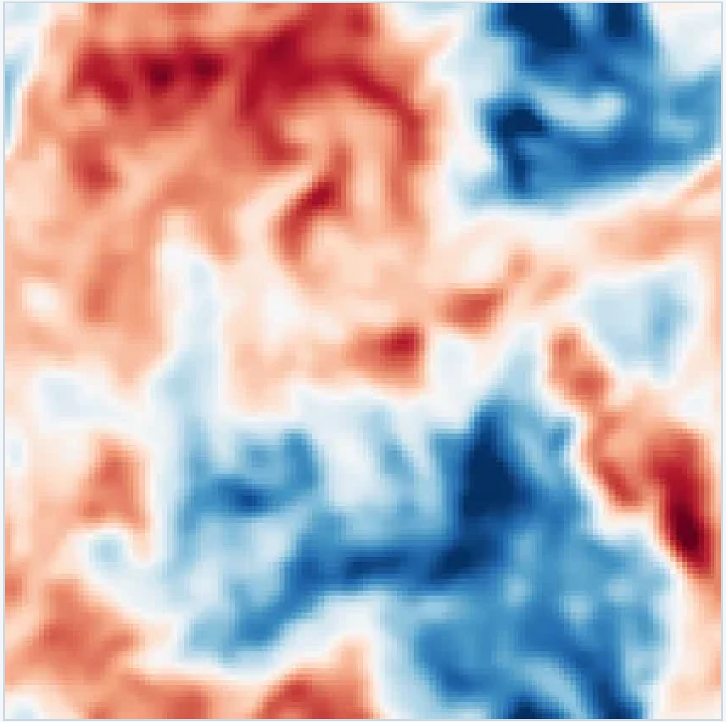
what an LES grid sees of a turbulent flow

The same flow, on two grids



DNS · VELOCITY u

810^3 — resolves everything

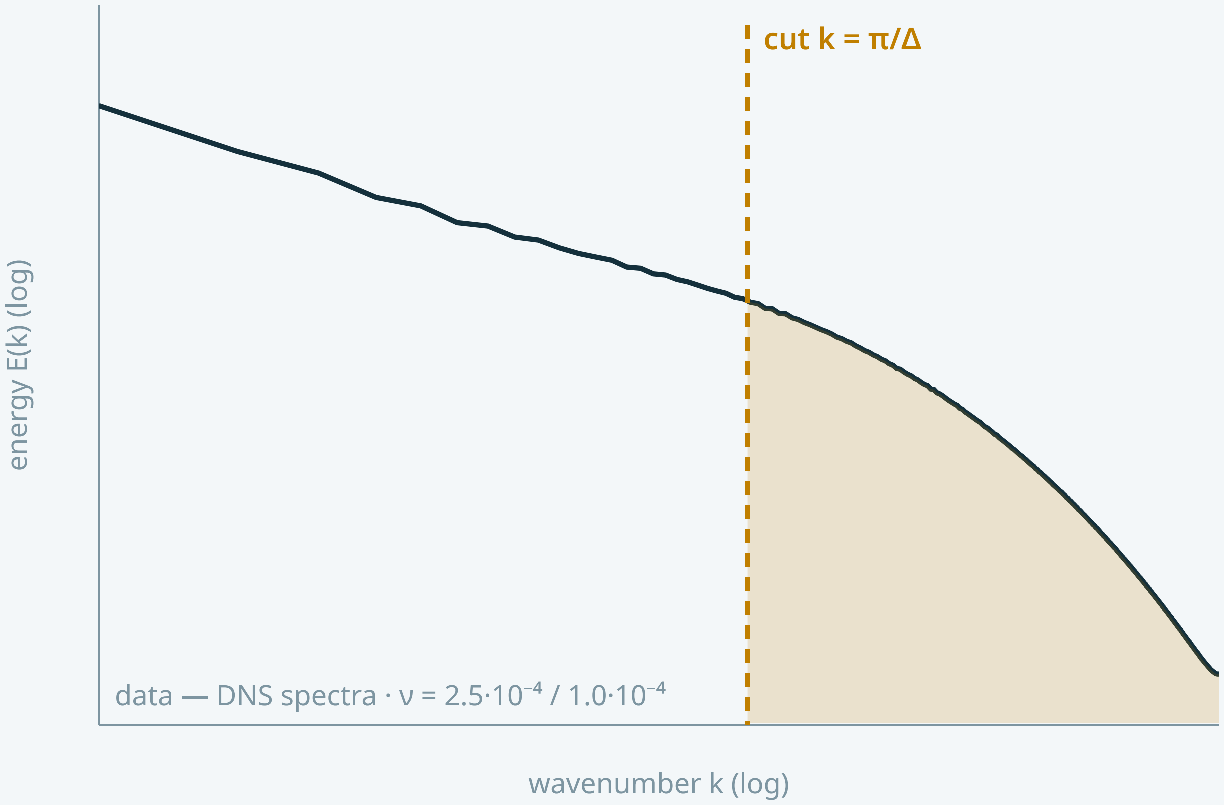


LES · FILTERED VELOCITY ū

128^3 — 250× fewer points, misses the small scales

the missing scales still act on the resolved ones — that action is the closure's job

Where the energy lives — and where the grid stops



the grid resolves everything left of the cut;
the shaded energy is invisible

coarsen the filter (Δ ↑) — more energy hidden

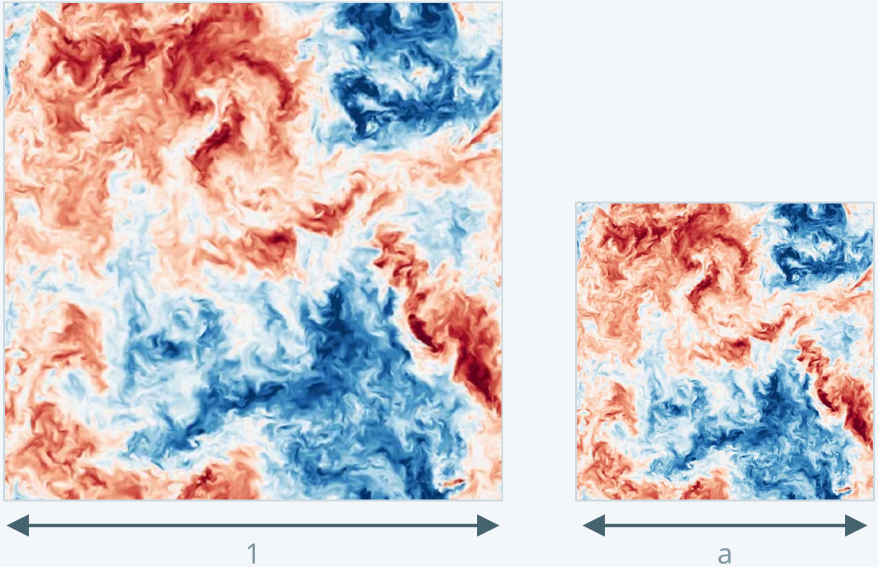
lower the viscosity (ν ↓) — the spectrum grows a longer tail

$$\mathrm{Re}_\Delta = \frac{\Delta^2 \, \|\bar{A}\|}{\nu}$$

one resolved-scale number for
how much lies beyond the cut

$\bar{A} = \nabla\bar{u}$ — the resolved velocity gradient · $\|\bar{A}\|$ its r.m.s. over the box

Why this number? Scale the equations and see what survives



a turbulent solution

…rescaled – still a solution

$$\nabla\cdot u = 0, \qquad \partial_t u + \nabla\cdot\left(u\,u^{\mathsf{T}} + p\,I - \nu\,\nabla u\right) = f$$

incompressible Navier-Stokes

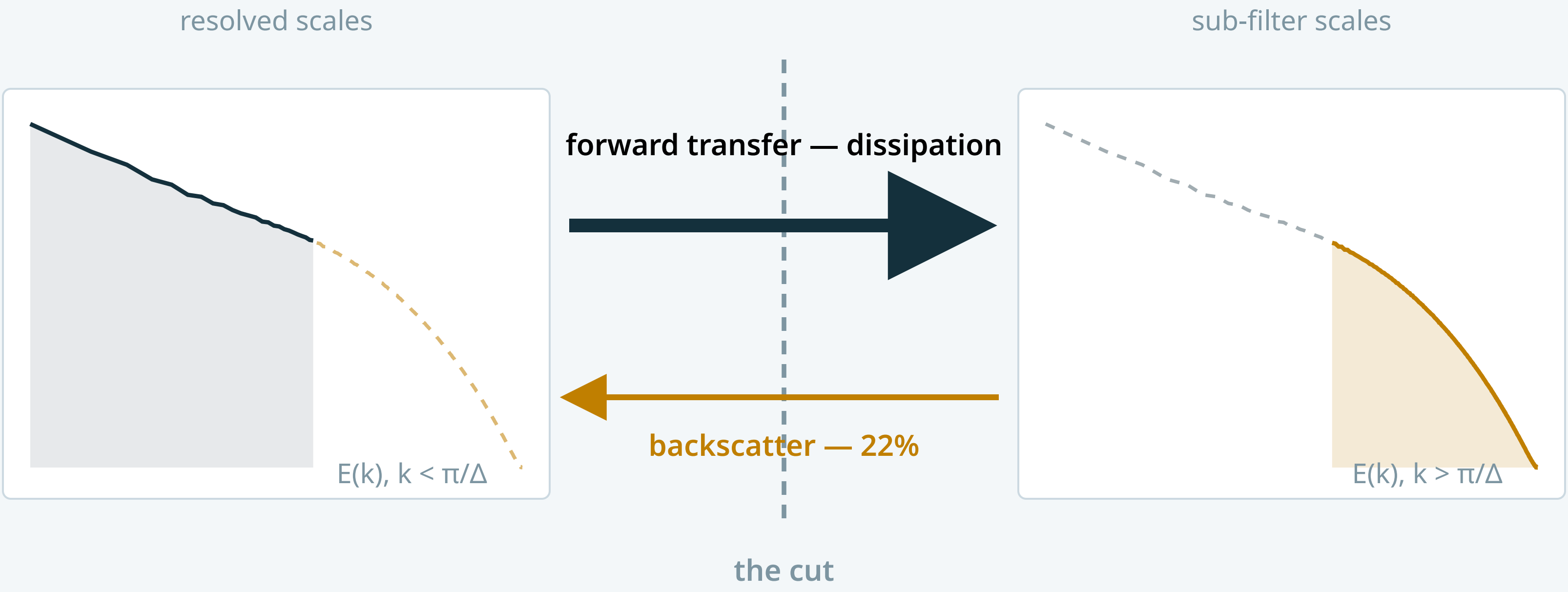
rescale space, time – and hence velocity:

$$x \to ax, \quad t \to bt, \quad u \to \tfrac{a}{b}\,u, \quad \nu \to \tfrac{a^2}{b}\,\nu$$

Still Navier–Stokes — but the rescaling multiplies ν by a^2/b. A closure blind to ν stays consistent only where $\mathbf{a^2 = b}$, the scalings that leave ν fixed.

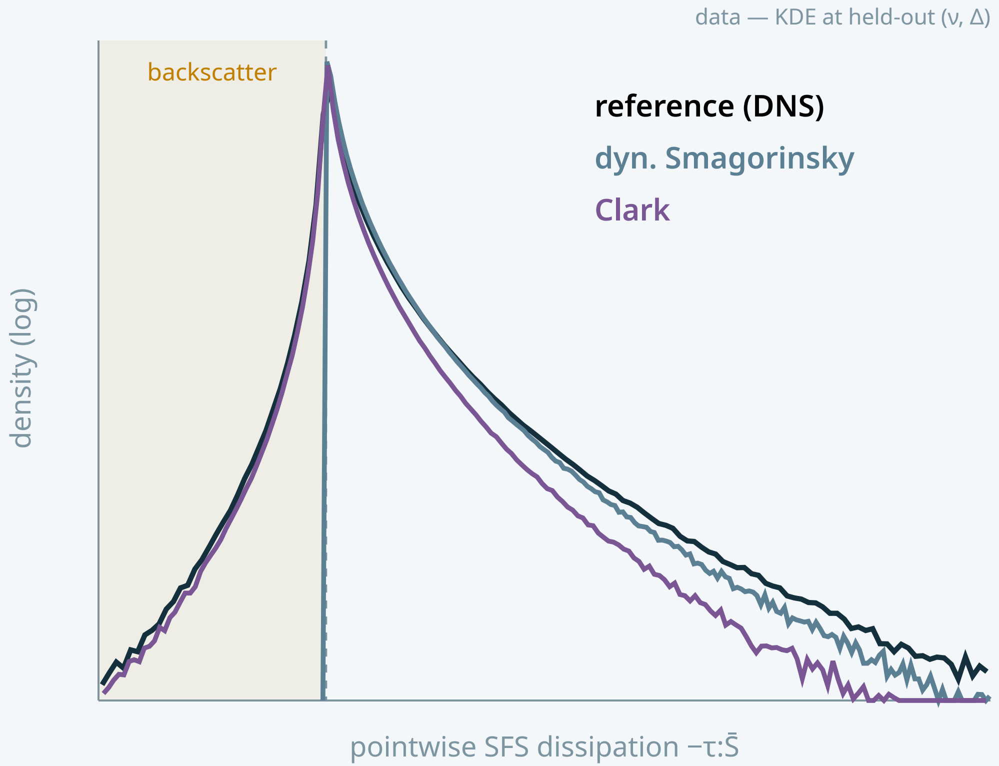
Along that family, $\bar{u}$ has a single dimensionless invariant: **Re_Δ**. Symmetry itself hands the closure its Reynolds input.

Energy crosses the cut in both directions



$\Pi = -\tau : \bar{S}$ a closure must get this net flux right — at every Reynolds number

Classical closures pick one side of this trade



FUNCTIONAL · SMAGORINSKY

$$\tau = -2\,(C_s\Delta)^2\,|\bar{S}|\,\bar{S}$$

dissipation ≈ right · structure wrong

`backscatter 0.0004 — can't flow backward`

STRUCTURAL · CLARK

$$\tau = \frac{\Delta^2}{12}\,\bar{A}\,\bar{A}^{\mathsf{T}}$$

structure ≈ right · dissipation 0.6× too weak

`backscatter 0.25 — close to the true 0.22`

and notice: **ν appears in neither formula** no ν

Smagorinsky 1963 · Germano et al. 1991 · Clark et al. 1979

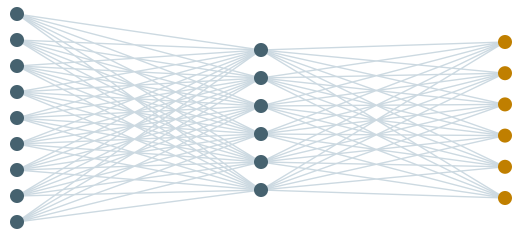
Or learn the closure from DNS — building in the physics we know

$$m : \underbrace{\bar{A}(x)}_{\mathbb{R}^{3\times 3}} \longmapsto \underbrace{\tau(x)}_{\text{sym. deviatoric}}$$

one network, fit a priori (least squares on τ) · 3 viscosities × 3 filter widths · 5 seeds

more inductive bias built in — less for the network to learn

MLP · NO SYMMETRY

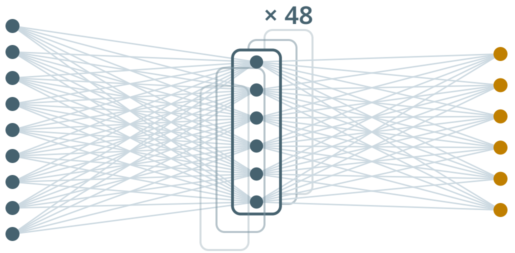


$\bar{A}$ (9) hidden τ (6)

9 gradients in → 6 stresses out

structure must be **learned** from data

G-CNN · TIED WEIGHTS



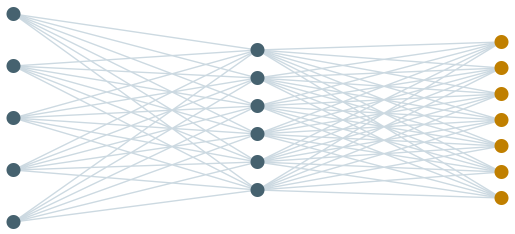
$\bar{A}$ (9) hidden τ (6)

one map, tied across 48 grid symmetries

equivariant by **weight sharing**

Cohen & Welling 2016

TBNN · TENSOR BASIS



λ (5) hidden α (7)

$$\tau = \sum_k \alpha_k(\lambda)\, T_k$$

λ = 5 invariants of $\bar{A}$ · $T_1 \ldots T_7$ fixed basis tensors

equivariant through the **features**

Ling et al. 2016

However much we build in, all three stay **single-point closures** — each reads τ(x) from $\bar{A}$(x) alone. Hold that thought.

The learned closures are just as blind

WITHOUT ν, THE SCALING SYMMETRY FORCES

$$\tau = \Delta^2 |\bar{A}|^2 \, m\Big(\bar{A}/|\bar{A}|\Big)$$ no ν

consistent — but **rigid three times over**:

τ can only grow as $\mathbf{\Delta^2}$

τ can only grow as $|\bar{A}|^2$

τ never feels **ν** — calibration frozen at the training regime

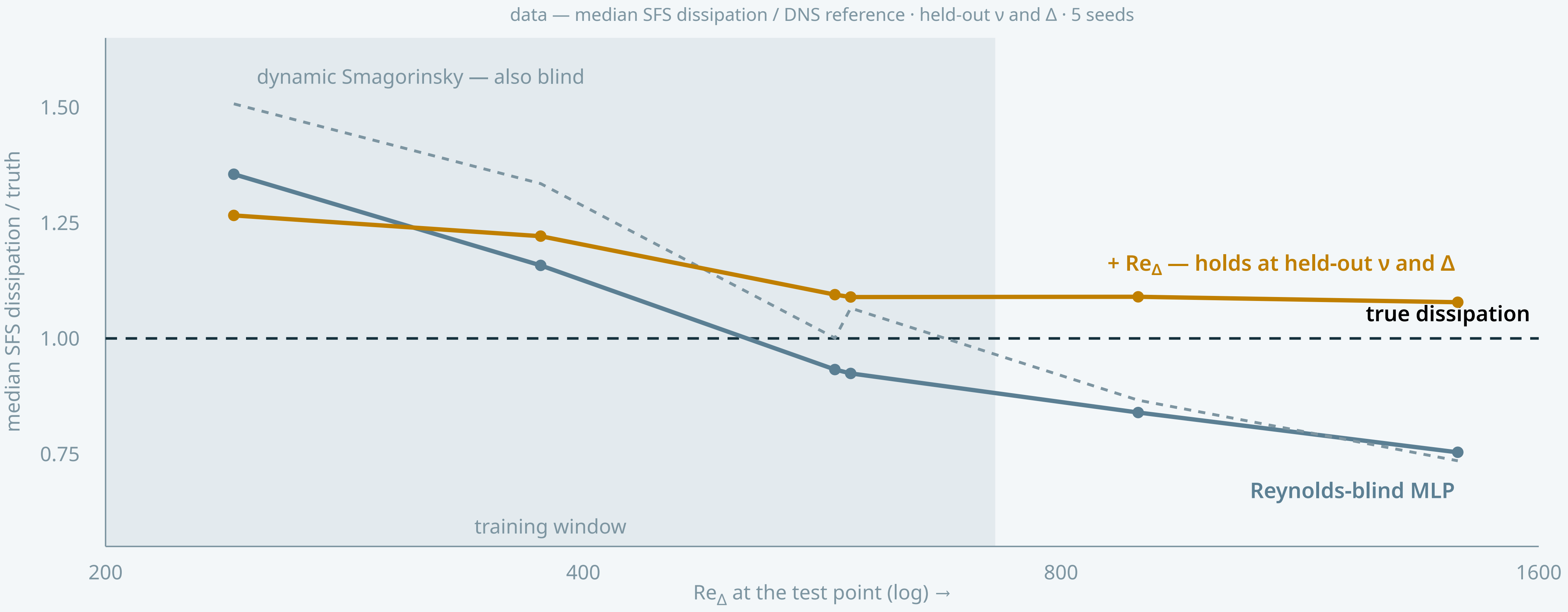
let ν back in — through the one dimensionless number the symmetry permits

SAME SYMMETRY, ONE EXTRA INPUT

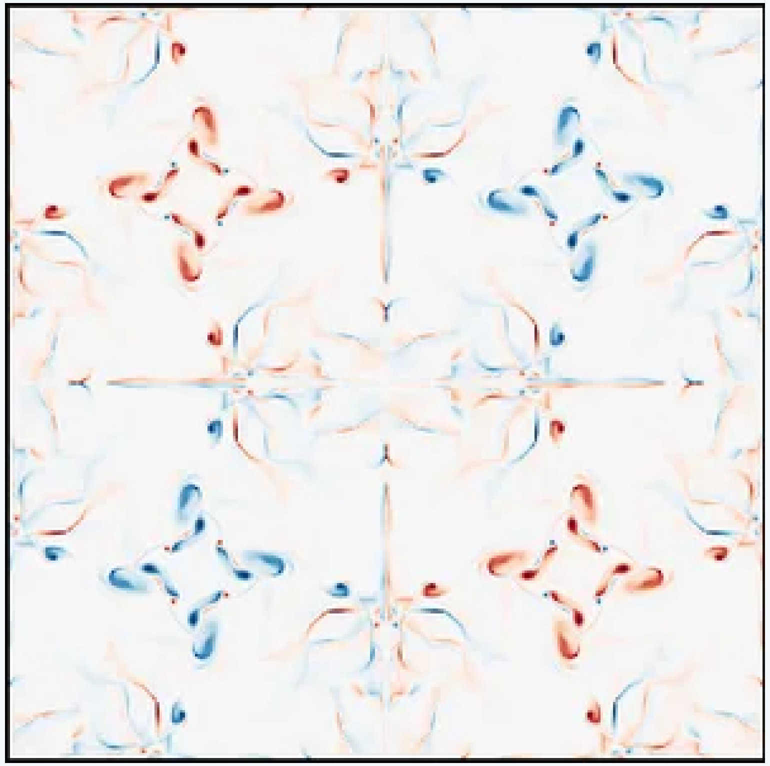
$$\tau = \Delta^2 |\bar{A}|^2 \, m\Big(\bar{A}/|\bar{A}|,\ \mathrm{Re}_\Delta\Big)$$ with ν

now m bends **nonlinearly** with Δ, |Ā| *and* ν —
but only through $\mathbf{Re_\Delta = \Delta^2 \|\bar{A}\|/\nu}$,
the one combination the rescalings leave alone

Tested away from training: who keeps their calibration?



A flow it has never seen: Taylor–Green becoming turbulent

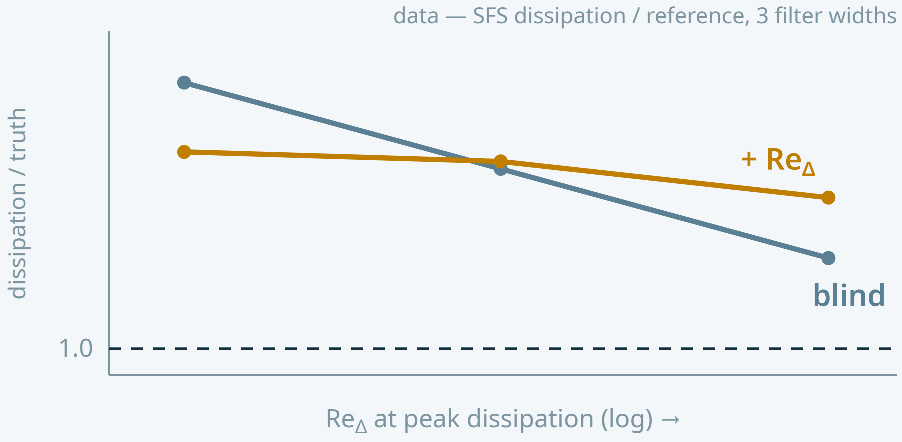


VORTICITY · $t = 0$

decaying, transitional — out of distribution twice over

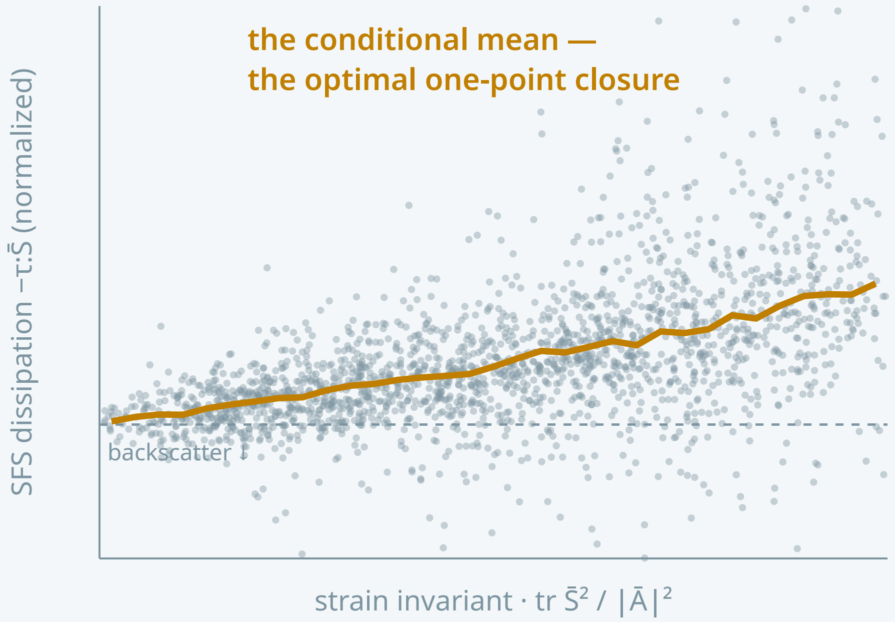
laminar → transition → decay

all closures stay stable — but over-dissipate

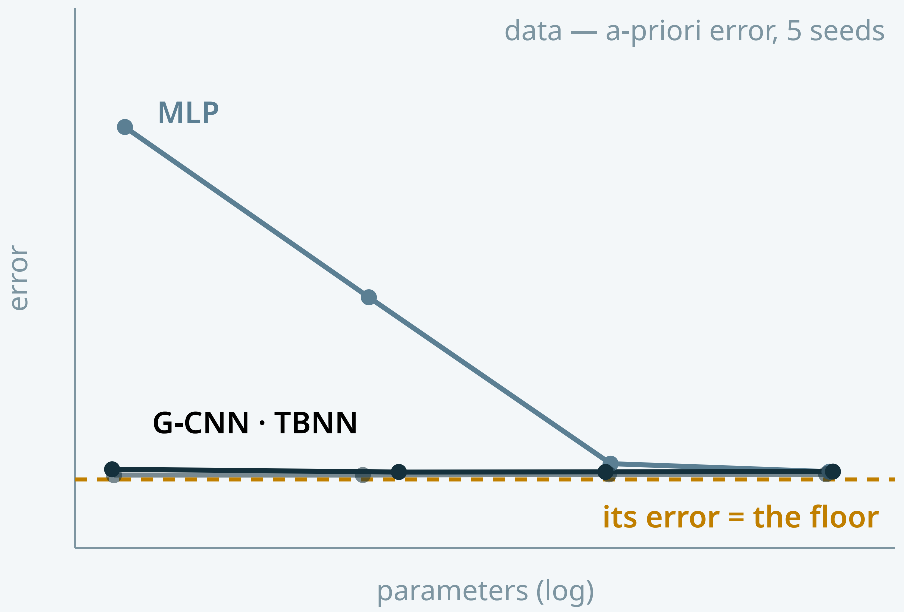


Re_Δ corrects the Reynolds-driven share of the excess, and honestly leaves the share owed to different flow physics

Couldn't a better architecture have fixed this? We checked.



real data — one slice · one input value ↦ a cloud of true stresses



every architecture saturates to the **same** one-point optimum — the equivariant ones with ~25× fewer parameters

so the Reynolds fix could never come from the architecture — **it had to come from the input**, available to all of them

TAKEAWAY

A closure that knows *what it's missing* works where it has never been.

Agdestein & Sanderse · CWI, Amsterdam · TU Eindhoven

“Approaching the optimal closure: equivariance, inductive bias, and Reynolds-number generalization in data-driven LES”

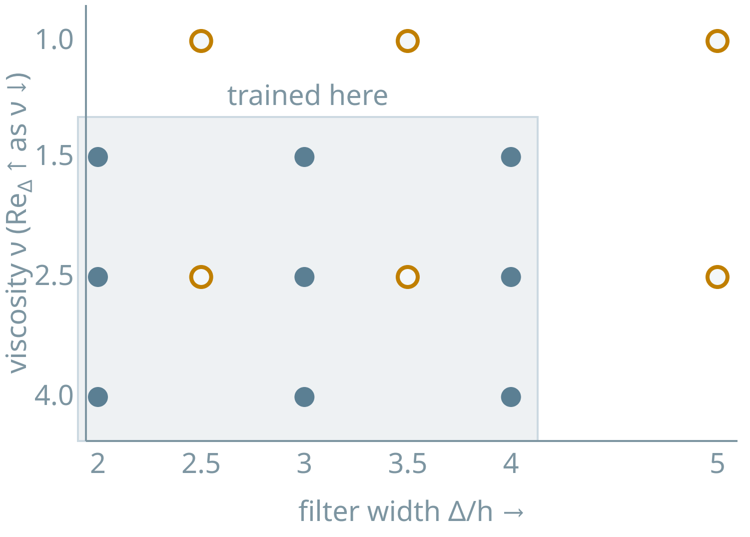
arxiv preprint: **2603.05325** · code on GitHub

Read the interactive story of this work
https://**agdestein**.github.io/

BACKUP

Setup: data and training

EXPERIMENT DESIGN



• trained (3×3) · ○ tested — held-out seed & ν, off-grid Δ

every test point sits off the training grid — in filter width *and* in Re_Δ

train $Re_\Delta \approx 104$–727 · 8 snapshots per grid point → 72 gradient-stress fields

DATA & TRAINING

DNS — forced HIT, pseudo-spectral 810^3 · one H100

2/3-rule dealiasing · $k_{max}\eta \geq 1.1$ · double precision

Filter — modified Gaussian onto a 128^3 LES grid

kernel commutes with the rescaling symmetry

Closures — MLP · G-CNN · TBNN, four sizes (≈120–3000 params)

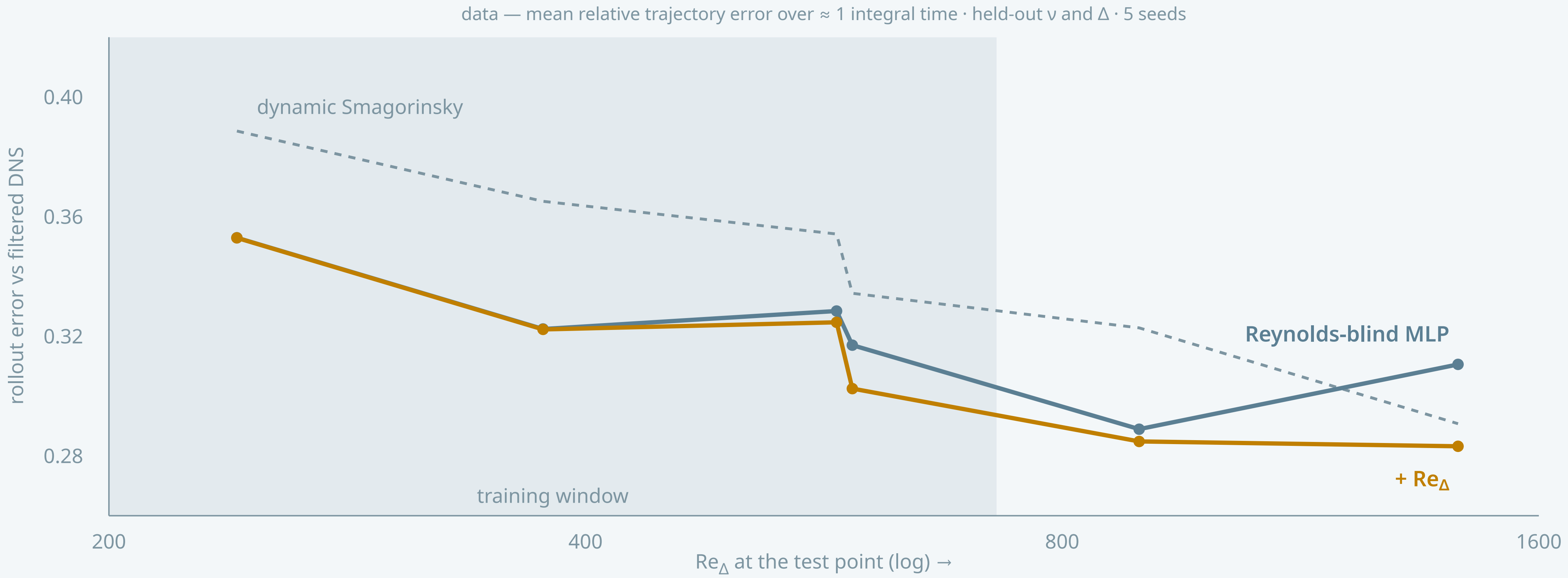
5 seeds each · a-priori least squares on the normalized stress · AdamW, 20 epochs

A posteriori — rollouts vs 40 filtered-DNS snapshots (≈1 integral time)

error = mean relative trajectory error · baselines: no-model · dynamic Smagorinsky · Clark

BACKUP

The calibration gain survives an actual LES run



inside the training window the two are essentially tied; the **Re_Δ input** pulls ahead at the highest Re (−9% at the farthest test point)

Scoreboard at the in-distribution test point (Δ/h = 2.5)

CLOSURE	A-PRIORI STRESS ERROR	A-POSTERIORI ROLLOUT ERROR
No-model	1.000	0.432
Dynamic Smagorinsky	0.964	0.389
Clark	0.466	0.364
MLP	0.449	0.353
G-CNN	0.449	0.355
TBNN	0.446	0.351
conditional-mean floor	**0.439**	**–**

a priori, the data-driven closures sit on the one-point floor; a posteriori the spread narrows but the ranking holds — the learned closures stay ahead of every classical baseline

learned closures: largest tier (≈3000 parameters), median over 5 seeds · all metrics at held-out seed $\nu = 2.5\cdot10^{-4}$, $\Delta/h = 2.5$